

## Solutions of Tutorial #2

### Topics covered:

- Topological spaces
- Bases and subbases
- The subspace topology

### Problem 1

For each  $x \in A$  we can choose an open set  $U_x$  containing  $x$  such that  $U_x \subset A$ . We then claim that  $\bigcup_{x \in A} U_x = A$ . We prove this by double inclusion.

First consider any  $y \in \bigcup_{x \in A} U_x$  so that there is an  $x \in A$  such that  $y \in U_x$ . Then clearly also  $y \in A$  since  $U_x \subset A$ . Hence  $\bigcup_{x \in A} U_x \subset A$  since  $y$  was arbitrary. Now consider  $y \in A$  so that clearly  $y \in U_y$ . Then obviously  $y \in \bigcup_{x \in A} U_x$  so that  $A \subset \bigcup_{x \in A} U_x$  since  $y$  was arbitrary. Thus we have shown that  $\bigcup_{x \in A} U_x = A$ , and since each  $U_x$  is open, it follows from the definition of a topology that the union  $\bigcup_{x \in A} U_x = A$  is open as well.

### Problem 2

- (i) We show that  $\mathcal{T}_\alpha$  is a topology on  $X$ . First, clearly  $\emptyset \in \mathcal{T}_\alpha$  since  $X - \emptyset = X$  is all of  $X$ . Also  $X \in \mathcal{T}_\alpha$  since  $X - X = \emptyset$  is countable. Now suppose that  $\mathcal{A}$  is a subcollection of  $\mathcal{T}_\alpha$  so that  $X - U$  is countable (or all of  $X$ ) for every  $U \in \mathcal{A}$ . Then we have that

$$X - \bigcup_{A \in \mathcal{A}} A = \bigcap_{A \in \mathcal{A}} (X - A)$$

is, by Corollary 7.3, countable (or all of  $X$ ) since every  $X - A$  is countable (or all of  $X$ ). Therefore  $\bigcup_{A \in \mathcal{A}} A \in \mathcal{T}_\alpha$  by definition.

Now suppose that  $U_1, \dots, U_n$  are nonempty elements of  $\mathcal{T}_\alpha$  so that  $X - U_i$  is a countable subset of  $X$  or  $X$  itself for each  $i \in \{1, \dots, n\}$ . Then we have

$$X - \bigcap_{i=1}^n U_i = \bigcup_{i=1}^n (X - U_i)$$

is a finite union of sets that are either countable subsets of  $X$ , or  $X$  itself. It then follows from Theorem 7.5 that the union is countable or  $X$  itself so that  $\bigcap_{i=1}^n U_i \in \mathcal{T}_\alpha$  by definition. This completes the proof that  $\mathcal{T}_\alpha$  is a topology on  $X$ .

- (ii) We claim that the collection  $\mathcal{T}_\infty$  as defined above is not always a topology on  $X$ . As a counterexample, let  $X = \mathbb{Z}_+$  and suppose that  $\mathcal{T}_\infty$  is a topology on  $X$ . Clearly if  $U$  is a finite subset of  $X$ , then  $X - U$  is infinite since  $X$  is infinite so that  $U$  is open. Now consider the subcollection

$$\mathcal{A} = \{\{i\} \mid i \in \mathbb{Z}_+ \text{ and } i > 1\} = \{\{2\}, \{3\}, \dots\}$$

Then clearly we have that  $\bigcup_{A \in \mathcal{A}} A = \{2, 3, \dots\}$  so that  $X - \bigcup_{A \in \mathcal{A}} A = \{1\}$  is neither infinite, empty, nor all of  $X$ . Therefore  $\bigcup_{A \in \mathcal{A}} A$  cannot be open, which violates property (2) of a topology. So it must be that  $\mathcal{T}_\infty$  is not a topology, which of course contradicts our assumption.

### Problem 3

- (i) First we show that  $\bigcap \mathcal{T}_\alpha$  is a topology on  $X$ . First, clearly since  $\emptyset$  and  $X$  are in every  $\mathcal{T}_\alpha$  since they are topologies, they are both in  $\bigcap \mathcal{T}_\alpha$  so that property (1). Now suppose that  $\mathcal{A}$  is a subcollection of  $\bigcap \mathcal{T}_\alpha$ . Consider any  $\mathcal{T}_\beta$  and any  $A \in \mathcal{A}$ . Then  $A$  is also in  $\bigcap \mathcal{T}_\alpha$  since  $\mathcal{A} \subset \bigcap \mathcal{T}_\alpha$ . It then follows that  $A$  is in our specific  $\mathcal{T}_\beta$ . Since  $A$  was arbitrary it follows that  $\mathcal{A}$  is a subcollection of  $\mathcal{T}_\beta$  so that  $\bigcup \mathcal{A} \in \mathcal{T}_\beta$  also since  $\mathcal{T}_\beta$  is a topology. Since  $\mathcal{T}_\beta$  was also arbitrary it follows that  $\bigcup \mathcal{A} \in \bigcap \mathcal{T}_\alpha$ . Lastly, since the subcollection  $\mathcal{A}$  was arbitrary, this shows property (2) for  $\bigcap \mathcal{T}_\alpha$ . Finally, suppose that  $U_1, \dots, U_n$  are sets in  $\bigcap \mathcal{T}_\alpha$ . Consider any  $\mathcal{T}_\beta$  so that clearly then  $U_i \in \mathcal{T}_\beta$  for every  $i \in \{1, \dots, n\}$ . It then follows that  $\bigcap_{i=1}^n U_i \in \mathcal{T}_\beta$  since  $\mathcal{T}_\beta$  is a topology. Since  $\mathcal{T}_\beta$  was arbitrary, this shows that  $\bigcap_{i=1}^n U_i \in \bigcap \mathcal{T}_\alpha$ , which shows property (3) for  $\bigcap \mathcal{T}_\alpha$ . This completes the proof that  $\bigcap \mathcal{T}_\alpha$  is a topology on  $X$  since all three properties have been shown.

Now we claim that  $\bigcup \mathcal{T}_\alpha$  is not generally a topology. Indeed, as a counterexample consider the set  $X = \{a, b, c\}$ , the topologies  $\mathcal{T}_1 = \{\emptyset, X, \{a\}\}$  and  $\mathcal{T}_2 = \{\emptyset, X, \{b\}\}$ , and the collection of topologies  $\mathcal{C} = \{\mathcal{T}_1, \mathcal{T}_2\}$ . Then we clearly have that  $\mathcal{T}_1 \cup \mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b\}\}$ , which is not a topology since  $\{\{a\}, \{b\}\}$  is a subcollection of  $\mathcal{T}_1 \cup \mathcal{T}_2$  but  $\{a\} \cup \{b\} = \{a, b\}$  is not in  $\mathcal{T}_1 \cup \mathcal{T}_2$ .

- (ii) First we show that there is a unique smallest topology that contains each  $\mathcal{T}_\alpha$ . It was proven in part (i) that  $\bigcup \mathcal{T}_\alpha$  is not necessarily a topology. However, it is clearly always a subbasis for a topology since clearly  $X \in \bigcup \mathcal{T}_\alpha$  since it is in each  $\mathcal{T}_\alpha$  since they are topologies. Hence obviously then  $\bigcup (\bigcup \mathcal{T}_\alpha) = X$  so that  $\bigcup \mathcal{T}_\alpha$  is a subbasis by definition. Then let  $\mathcal{T}_s$  be the topology generated by the subbasis  $\bigcup \mathcal{T}_\alpha$ . We claim that  $\mathcal{T}_s$  is then the smallest topology that contains all the  $\mathcal{T}_\alpha$  as subsets.

First, from the proof following the definition of a subbasis, we know that the set  $\mathcal{B}$  of finite intersections of elements of  $\bigcup \mathcal{T}_\alpha$  is a basis for the topology  $\mathcal{T}_s$ , and that  $\mathcal{T}_s$  is the set of all unions of subcollections of  $\mathcal{B}$ . We first show that every  $\mathcal{T}_\alpha$  is indeed contained as a subset of  $\mathcal{T}_s$ . So consider any specific  $\mathcal{T}_\beta$  and any  $U \in \mathcal{T}_\beta$ . Then clearly  $U \in \bigcup \mathcal{T}_\alpha$

so that  $U \in \mathcal{B}$  since  $U = \bigcap \{U\}$  is a finite intersection of elements of  $\bigcup \mathcal{T}_\alpha$ . It then follows that  $U \in \mathcal{T}_s$  since  $U = \bigcup \{U\}$  is the union of a subcollection of  $\mathcal{B}$ . Since  $U$  was arbitrary, this shows that  $\mathcal{T}_\beta \subset \mathcal{T}_s$ , which shows the result since  $\mathcal{T}_\beta$  was arbitrary.

Now we show that  $\mathcal{T}_s$  is the smallest such topology as ordered by  $\subsetneq$ . So suppose that  $\mathcal{T}$  is a topology that contains every  $\mathcal{T}_\alpha$  as a subset. Consider any  $U \in \mathcal{T}_s$  so that  $U = \bigcup \mathcal{C}$  for some subcollection  $\mathcal{C} \subset \mathcal{B}$ . Now consider any  $Y \in \mathcal{C}$  so that also  $Y \in \mathcal{B}$ . Then  $Y = \bigcap_{i=1}^n Y_i$  where each  $Y_i \in \bigcup \mathcal{T}_\alpha$ . Then each  $Y_i$  is in some  $\mathcal{T}_\beta \subset \mathcal{T}$  so that also  $Y_i \in \mathcal{T}$ . Since  $\mathcal{T}$  is a topology, it follows that the finite intersection  $\bigcap_{i=1}^n Y_i = Y$  is also in  $\mathcal{T}$ . Since  $Y$  was arbitrary, this shows that  $\mathcal{C} \subset \mathcal{T}$  so that  $\mathcal{C}$  is a subcollection of  $\mathcal{T}$ . It then follows that  $\bigcup \mathcal{C} = U$  is also in  $\mathcal{T}$  since  $\mathcal{T}$  is a topology. Since  $U$  was arbitrary, we have that  $\mathcal{T}_s \subset \mathcal{T}$ , which shows that  $\mathcal{T}_s$  is the smallest topology since  $\mathcal{T}$  was arbitrary. It is easy to see that  $\mathcal{T}_s$  is unique since, if both  $\mathcal{T}_1$  and  $\mathcal{T}_s$  are the smallest topologies that contain each  $\mathcal{T}_\alpha$  as subsets, then we would have that both  $\mathcal{T}_1 \subset \mathcal{T}_2$  and  $\mathcal{T}_2 \subset \mathcal{T}_1$  so that  $\mathcal{T}_1 = \mathcal{T}_2$ . Really this follows from the more general fact that smallest elements in any order are always unique.

Next we show that there is a unique largest topology that is contained in each  $\mathcal{T}_\alpha$ . It was shown in part (i) that  $\mathcal{T}_l = \bigcap \mathcal{T}_\alpha$  is a topology on  $X$ . We claim that in fact this is the unique largest topology contained in all  $\mathcal{T}_\alpha$ . First, clearly  $\mathcal{T}_l = \bigcap \mathcal{T}_\alpha$  is contained in each  $\mathcal{T}_\alpha$  since the intersection of a collection of sets is always a subset of every set in the collection. Now suppose that  $\mathcal{T}$  is a topology that is contained in every  $\mathcal{T}_\alpha$ , i.e.  $\mathcal{T} \subset \mathcal{T}_\alpha$  for every  $\mathcal{T}_\alpha$ . Then clearly for any  $U \in \mathcal{T}$  we have that  $U \in \mathcal{T}_\alpha$  for every  $\mathcal{T}_\alpha$  so that  $U \in \bigcap \mathcal{T}_\alpha = \mathcal{T}_l$ . Thus  $\mathcal{T} \subset \mathcal{T}_l$  since  $U$  was arbitrary. This shows that  $\mathcal{T}_l$  is the largest such topology since  $\mathcal{T}$  was arbitrary. Clearly also  $\mathcal{T}_l$  is unique since, if  $\mathcal{T}_1$  and  $\mathcal{T}_2$  are two such largest topologies that are contained in every  $\mathcal{T}_\alpha$ . Then we would have  $\mathcal{T}_1 \subset \mathcal{T}_2$  and  $\mathcal{T}_2 \subset \mathcal{T}_1$  so that  $\mathcal{T}_1 = \mathcal{T}_2$ . This also follows from the fact that the largest element in any ordered set (or collection of sets in this case) is unique.

- (iii) Note that the proofs in part (ii) are constructive so that we can construct these topologies as done in the proof. For the smallest topology containing  $\mathcal{T}_1$  and  $\mathcal{T}_2$  we have that

$$\bigcup \{\mathcal{T}_1, \mathcal{T}_2\} = \mathcal{T}_1 \cup \mathcal{T}_2 = \{\emptyset, X, \{a\}, \{a, b\}, \{b, c\}\}$$

is a subbasis for the smallest topology  $\mathcal{T}_s$ . Then the collection of all finite intersections of elements of this set is a basis for  $\mathcal{T}_s$ :

$$\mathcal{B} = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}, \{b, c\}\}$$

Then the topology  $\mathcal{T}_s$  is the set of all unions of subcollections of  $\mathcal{B}$ :

$$\mathcal{T}_s = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}, \{b, c\}\} = \mathcal{B}$$

so that evidently the basis and the topology are the same set here! For the largest topology contained in  $\mathcal{T}_1$  and  $\mathcal{T}_2$  we have simply

$$\mathcal{T}_l = \bigcap \{\mathcal{T}_1, \mathcal{T}_2\} = \mathcal{T}_1 \cap \mathcal{T}_2 = \{\emptyset, X, \{a\}\}.$$

#### Problem 4

Suppose that  $\mathcal{T}$  is the topology generated by basis  $\mathcal{A}$ , and  $\mathcal{C}$  is the collection of topologies on  $X$  that contain  $\mathcal{A}$  as a subset. First we show that  $\mathcal{T} = \bigcap_{\mathcal{T}_c \in \mathcal{C}} \mathcal{T}_c$ . Consider  $U \in \mathcal{T}$  and any  $\mathcal{T}_c \in \mathcal{C}$  so that  $\mathcal{A} \subset \mathcal{T}_c$ . Then, since  $\mathcal{A}$  generates  $\mathcal{T}$ , it follows from Lemma 13.1 that  $U$  is the union of elements of  $\mathcal{A}$ . Clearly then each of these elements of  $\mathcal{A}$  is in  $\mathcal{T}_c$  since  $\mathcal{A} \subset \mathcal{T}_c$  so that their union is as well since  $\mathcal{T}_c$  is a topology. Hence  $U \in \mathcal{T}_c$  so that  $\mathcal{T} \subset \mathcal{T}_c$  since  $U$  was arbitrary. Hence  $\mathcal{T}$  is contained in all elements of  $\mathcal{C}$  so that  $\mathcal{T} \subset \bigcap_{\mathcal{T}_c \in \mathcal{C}} \mathcal{T}_c$ . Also, clearly  $\mathcal{T}$  is a topology that contains  $\mathcal{A}$  so that  $\mathcal{T} \in \mathcal{C}$ . Clearly then  $\bigcap_{\mathcal{T}_c \in \mathcal{C}} \mathcal{T}_c \subset \mathcal{T}$  so that  $\mathcal{T} = \bigcap_{\mathcal{T}_c \in \mathcal{C}} \mathcal{T}_c$  as desired.

Next we show the same thing but when  $\mathcal{A}$  is a subbasis. Let  $\mathcal{B}$  be the set of all finite intersections of elements of  $\mathcal{A}$ , which we know is a basis for  $\mathcal{T}$  by the proof after the definition of a subbasis. We show that  $\mathcal{B} \subset \mathcal{T}_c$  for all  $\mathcal{T}_c \in \mathcal{C}$ . So consider any set  $B \in \mathcal{B}$  so that  $B$  is the finite intersection of elements of  $\mathcal{A}$ . Also consider any  $\mathcal{T}_c \in \mathcal{C}$  so that each of these elements is in  $\mathcal{T}_c$  since  $\mathcal{A} \subset \mathcal{T}_c$ . Since  $\mathcal{T}_c$  is a topology, clearly the finite intersection of these elements, i.e.  $B$ , is in  $\mathcal{T}_c$ . Hence  $\mathcal{B} \subset \mathcal{T}_c$  since  $B$  was arbitrary. It then follows from what was shown before that  $\mathcal{T} = \bigcap_{\mathcal{T}_c \in \mathcal{C}} \mathcal{T}_c$  since  $\mathcal{T}$  is the topology generated by the basis  $\mathcal{B}$  and  $\mathcal{B}$  is contained in each topology in  $\mathcal{C}$ .

#### Problem 5

Let  $\mathcal{T}$  be the standard topology on  $\mathbb{R}$ . First, clearly  $\mathcal{B}$  is a collection of open sets of  $\mathcal{T}$  since each element is a basis element in the standard basis (i.e. an open interval). Now consider any  $U \in \mathcal{T}$  and any  $x \in U$ . Then there is a standard basis element  $B' = (a', b')$  such that  $x \in B'$  and  $B' \subset U$  since  $\mathcal{T}$  is generated by the standard basis. Then  $a' < x < b'$  so that, since the rationals are order-dense in the reals (see Exercise 4.9 part (d)), there are rational  $a$  and  $b$  such that  $a' < a < x < b < b'$ . Let  $B = (a, b)$  so that clearly  $x \in B, B \subset B' \subset U$ , and  $B \in \mathcal{B}$ . This shows that  $\mathcal{B}$  is a basis for  $\mathcal{T}$  by Lemma 13.2 since  $U$  and  $x \in U$  were arbitrary.

#### Problem 6

Let  $\mathcal{T}$  be the topology on  $X$  and  $\mathcal{T}_Y$  be the subspace topology that  $Y$  inherits from  $X$ . Also let  $\mathcal{T}_A$  and  $\mathcal{T}'_A$  be the topologies that  $A$  inherits as a subspace of  $Y$  and  $X$ , respectively. Therefore we must show that  $\mathcal{T}_A = \mathcal{T}'_A$ . Now, by definition of subspace topologies we have that,

$$\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}\} \quad \mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}_Y\} \quad \mathcal{T}'_A = \{A \cap U \mid U \in \mathcal{T}\}.$$

Now suppose that  $W \in \mathcal{T}_A$  so that  $W = A \cap V$  for some  $V \in \mathcal{T}_Y$ . Then we have that  $V = Y \cap U$  for some  $U \in \mathcal{T}$ , and hence

$$W = A \cap V = A \cap (Y \cap U) = (A \cap Y) \cap U = A \cap U$$

since we have that  $A \cap Y = A$  since  $A \subset Y$ . Since  $U \in \mathcal{T}$  this clearly shows that  $W \in \mathcal{T}'_A$  so that  $\mathcal{T}_A \subset \mathcal{T}'_A$  since  $W$  was arbitrary. Then, for any  $W \in \mathcal{T}'_A$ , we have that  $W = A \cap U$  for some  $U \in \mathcal{T}$ . Let  $V = Y \cap U$  so that clearly  $V \in \mathcal{T}_Y$ . Then as before we have that  $A = A \cap Y$  since  $A \subset Y$  so that

$$W = A \cap U = (A \cap Y) \cap U = A \cap (Y \cap U) = A \cap V,$$

and thus  $W \in \mathcal{T}_A$  since  $V \in \mathcal{T}_Y$ . Since  $W$  was arbitrary this shows that  $\mathcal{T}'_A \subset \mathcal{T}_A$ , which completes the proof that  $\mathcal{T}_A = \mathcal{T}'_A$ .